

ANKITA JAIN

+1 704-904-6822 • ankita.utd@gmail.com • linkedin.com/in/jainankita

SUMMARY

Principal AI & Data Platform Engineer with 12+ years of experience architecting scalable, cloud-native distributed systems, enterprise data platforms, and production-grade Generative AI environments for consumer and enterprise products. Proven technical authority who balances keyboard-level depth (Python, PySpark, Scala) with executive-level technical storytelling to drive cross-organizational initiatives. Expert in engineering reusable AI patterns and agentic workflows, LLM evaluation platforms, and robust Responsible AI/observability frameworks within highly scaled, regulated enterprise environments.

EXPERIENCE

Microsoft (D365 Sales AI) | Charlotte, NC

Principal AI & Data Platform Engineer (Lead)

Mar 2020 – Present

- **Enterprise AI Strategy & Evaluation:** Architected and engineered a reusable, generic enterprise LLM evaluation platform (rubrics, side-by-side testing, automated scoring, real-time dashboards) scaled across 30+ cross-functional teams to gate production releases and track regressions.
- **Agentic Systems & Prototyping:** Hands-on designer and builder of production-grade agentic AI systems utilizing Copilot Studio and Agent Builder; successfully launched multiple autonomous agents to GA and executed 20+ complex POCs balancing quality, latency, safety, and cost.
- **Responsible AI & Governance:** Implemented stringent Responsible AI guardrails for enterprise applications, including real-time policy enforcement, jailbreak defenses, cross-domain grounding, and end-to-end observability metrics (token cost, latency percentiles, error budgets).
- **Production SLO Optimization:** Established and operated rigorous production SLOs (p95 latency, cost, quality) for customer-facing AI agents, consistently driving metrics to ≥95% groundedness and ≥90% usefulness.
- **Advanced AI Engineering:** Developed production-ready Dynamics 365 Sales AI agents, MCP-powered tools, Claude skills, and RAG-based assistants, reducing manual corporate lookup times by 75%.
- **Technical Authority & Leadership:** Serve as trusted advisor to senior leadership, guiding a team of 6 engineers to embed deep automated evaluations into the continuous build-measure-learn product loop.
- **Cloud-Native Architecture:** Architected orchestration microservices and bulk asset enrichment pipelines for Azure Purview, increasing data throughput by 50% across 300M+ enterprise assets utilizing Kubernetes and REST microservices.
- **Data Platform Engineering:** Designed and scaled Azure Purview Cold Tier analytics processing pipelines using PySpark and developed Cosmos DB snapshot synchronization frameworks, substantially dropping compute and storage overhead by 50%.

Bank of America (Enterprise Credit Risk) | Charlotte, NC

Senior Data Engineer

Feb 2016 – Mar 2020

- **Reusable Framework Innovation:** Conceptualized, coded, and open-sourced an internal reusable Spark-based ETL framework adopted by 25 distinct internal business lines across regulated financial domains, reducing manual reporting workloads by 65% enterprise-wide while maintaining compliance with financial data governance standards.
- **Legacy Modernization:** Spearheaded the technical migration of legacy Netezza ETL pipelines to Hive, Impala, and Spark within a regulated financial services environment; optimized complex SQL queries using Scala Spark, slashing core ETL processing time by 50% while preserving audit trail and compliance requirements.

- **Technical Storytelling & Mentorship:** Acted as technical authority and educator across the organization, authoring architecture playbooks and training over 50 associates across 5 distinct teams to scale Hadoop/Spark adoption.
- **Risk Data Aggregation:** Engineered Spark-SQL data pipelines to securely aggregate Wholesale Credit Risk customer and corporate loan data, ensuring integrity and traceability for downstream regulatory compliance reporting across enterprise credit risk governance frameworks.

Tresata, Inc. | Charlotte, NC

Solution / Data Science Engineer

Sep 2015 – Feb 2016

- Utilized Spark API, Scala, and Scalding to parse unstructured transaction logs and scale predictive recommendation workflows for high-volume consumer data portfolios.

TECHNICAL SKILLS

AI & GenAI: LLM Evaluation & Observability, Agentic Workflows, Prompt Engineering, Model Metrics Benchmarking, RAG Systems, Responsible AI Guardrails

Programming Languages: Python, PyTorch, Java, Scala, C#, SQL

Distributed Systems & Data Platforms: Spark, PySpark, Hadoop, Hive, Impala, Airflow, Cosmos DB, Azure Purview, Event Streaming

Cloud & Architecture: Azure, Cloudera, Cloud-Native Microservices, Infrastructure as Code (ARM Templates), Data Modeling, CI/CD Pipelines

EDUCATION

University of North Carolina | Charlotte, NC

Master of Science in Data Science | GPA: 4.0 | Aug 2015